Sa’eed Telvari

PhD Candidate in Petroleum Engineering

Edinburgh, UK st4014@hw.ac.uk ORCID 0000-0002-4896-295X Google Scholar

linkedin.com/in/stelvari saeedtelvari.github.io

Research interests

Reservoir simulation, CO_2 storage, vertical-equilibrium models, multiphase and compositional flow, machine learning for upscaling and digital rock, fractured reservoirs.

Education

2024 – present

PhD in Petroleum Engineering

Heriot-Watt University, Edinburgh

Thesis: Developing Vertical Equilibrium Models for Simulating CO_2 Storage in Depleted Gas Reservoirs

2022 – 2024

MSc in Petroleum Engineering (Reservoir)

Amirkabir University of Technology, Tehran

GPA 3.65/4 (17.23/20)

Thesis: Machine Learning Methods in Upscaling Fine-scale Discrete Fracture Models

2018 – 2022

BSc in Petroleum Engineering

Amirkabir University of Technology, Tehran

GPA 17.43/20

Thesis: Prediction of two-phase flow properties for digital sandstones using 3D CNNs

Research projects and experience

2024 – present

Doctoral research: vertical-equilibrium modelling

Institute of GeoEnergy Engineering, Heriot-Watt University

Reduced-order models of CO_2–methane–brine migration in depleted gas reservoirs, for screening, uncertainty analysis and field-scale forecasting.

2026

MuPSI: Multiscale Pressure-Stress Impacts on fault integrity for multi-site regional CO_2 storage

Clean Energy Transition Partnership

VE models for a field-scale case study of depleted gas fields, with NORCE, TU Delft and IDAEA-CSIC.

3D fault-detection tool for MRST

MuPhi research group, Heriot-Watt University

Extended a 2D code to 3D; it finds fault faces with signed distance fields on Cartesian and unstructured grids and is used by members of the MuPhi group.

2022 – 2026

Machine-learning permeability upscaling

Amirkabir University of Technology

CNN workflows that estimate effective permeability from heterogeneous fine-scale reservoir models.

2021 – 2023

Two-phase flow prediction for digital sandstones

Amirkabir University of Technology

3D convolutional neural networks that predict relative permeability and capillary pressure from digital-rock images.

Publications

2026

A Vertical Equilibrium Model for CO_2 Migration in Depleted Gas Fields

Telvari, S., Ramachandran, H., Wang, G., Doster, F.

EarthArXiv preprint. Under review in Geoenergy Science and Engineering. DOI **10.31223/X5P49D**

2026

Accelerated Permeability Upscaling: A Convolutional Neural Network Approach

Sayyafzadeh, M., **Telvari, S.**, Guérillot, D., Sharifi, M.

SPE Journal 31(04), 2242–2260. Journal article. DOI **10.2118/231836-PA**

2025

Three-Phase VE Simulation of CO_2–Methane–Brine Flow in Reservoirs

Telvari, S., Ramachandran, H., Wang, G., Doster, F.

Sixth EAGE Global Energy Transition Conference & Exhibition (GET 2025), Rotterdam. Extended abstract and poster. DOI **10.3997/2214-4609.202521145**

2023

Prediction of two-phase flow properties for digital sandstones using 3D convolutional neural networks

Telvari, S., Sayyafzadeh, M., Siavashi, J., Sharifi, M.

Advances in Water Resources 176, 104442. Journal article. DOI **10.1016/j.advwatres.2023.104442**

Presentations

Sep 2026

A Compositional Vertical Equilibrium Model for CO_2 Storage in Depleted Gas Reservoirs

Oral, 9th InterPore UK Chapter Conference, Edinburgh

May 2026

Vertical-Equilibrium Modelling of CO_2 Migration in Depleted Reservoirs

Oral, InterPore2026, 18th Annual Meeting, Nantes

Feb 2026

A Vertical Equilibrium Model for CO_2 Migration in Depleted Gas Fields

Seminar talk, IGE GeoEnergy Seminar Series, Heriot-Watt University

Oct 2025

Three-Phase VE Simulation of CO_2–Methane–Brine Flow in Reservoirs

Poster, Sixth EAGE Global Energy Transition Conference (GET 2025), Rotterdam

Teaching

2026 – 2027

Approved Demonstrator, MSc Petroleum Engineering

Heriot-Watt University, Edinburgh

Formation Evaluation labs (Nov 2026); Production Technology PipeSim tutorials (Jan – Feb 2027)

Feb – Apr 2026

Co-organiser and lead instructor, MATLAB/MRST workshop series

SET-IT, Heriot-Watt University

Led “Simulator Development with MRST’s Rapid Prototyping Framework” for IGE PhD students and postdocs

MSc

Teaching assistant

Amirkabir University of Technology, Tehran

Advanced Naturally Fractured Reservoirs; Advanced Enhanced Oil Recovery (Eclipse and Petrel tutorials)

Service and leadership

Sep 2026

Session co-chair, 9th InterPore UK Chapter Conference

Session 1: Multiphase Phenomena, with Dr Sarah Perez, Edinburgh

2026

Co-reviewer, Geophysical Research Letters

Manuscript review with Prof. Florian Doster

May 2026

Conference volunteer, InterPore2026

Technical session support for minisymposia MS05 and MS08, Nantes

2025

Webmaster, SPE Heriot-Watt University Student Chapter

SPE officer certificate of appreciation; the chapter won the 2026 SPE Regional Outstanding Student Chapter Award

Web contributor

SET-IT research community and MuPhi Research Group websites

Professional training

Sep 2026

SLB Intersect CCS training

Five-day course

Mar 2026

CMG Carbon Capture Underground Storage Modelling

Certificate of achievement, CMG CO_2 storage workshop

Skills

Programming	Python, MATLAB, Julia, Rust, LaTeX
Machine learning	TensorFlow, PyTorch, scikit-learn, OpenCV, OpenPNM
Reservoir simulation	MRST, Eclipse, Intersect, CMG, Petrel RE, PipeSim, Saphir, PVTSim
CFD and imaging	OpenFOAM, PerGeos, SALOME, MeshLab
Tools	Linux, Docker, Git, Jupyter, VS Code

Recognition

`2024 – present` **James Watt Scholarship**
Full PhD funding, Heriot-Watt University

`2026` **Runner-up, SPE Aberdeen Section Student Bursary**
£1,000 awarded

`2026` **Team runner-up, EAGE “The Energy–AI Nexus” Hackathon**
EAGE Annual 2026, Aberdeen

National undergraduate scholarship
Full tuition waiver